



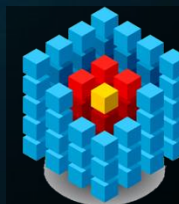
Inteligencia Artificial y Aprendizaje Automático

Sesión 5: Redes Neuronales
Convolucionales y Aplicaciones

Presentan:

Dr. Ulises Olivares Pinto

Fernando Ramírez González

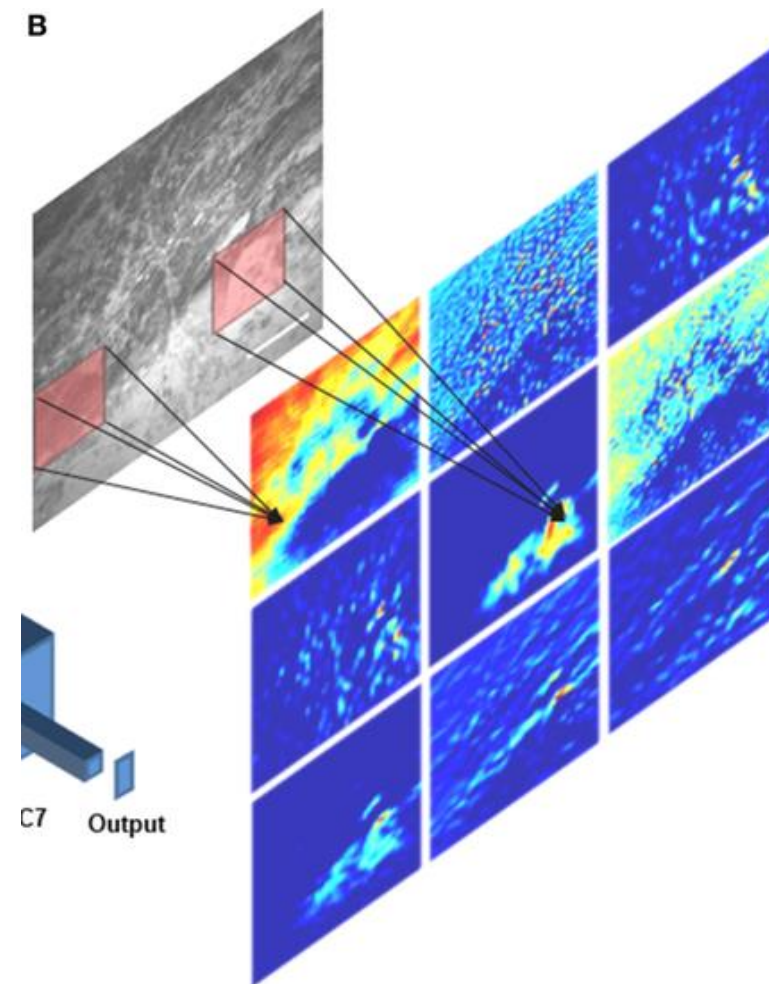


HPC
DATA LAB

Contenido

Redes Neuronales convolucionales

- Convolución
- Diseño de arquitecturas
- Ejercicios



Teachable Machine

<https://teachablemachine.withgoogle.com/>

New Project

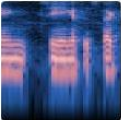
Open an existing project from Drive.

Open an existing project from a file.



Image Project

Teach based on images, from files or your webcam.



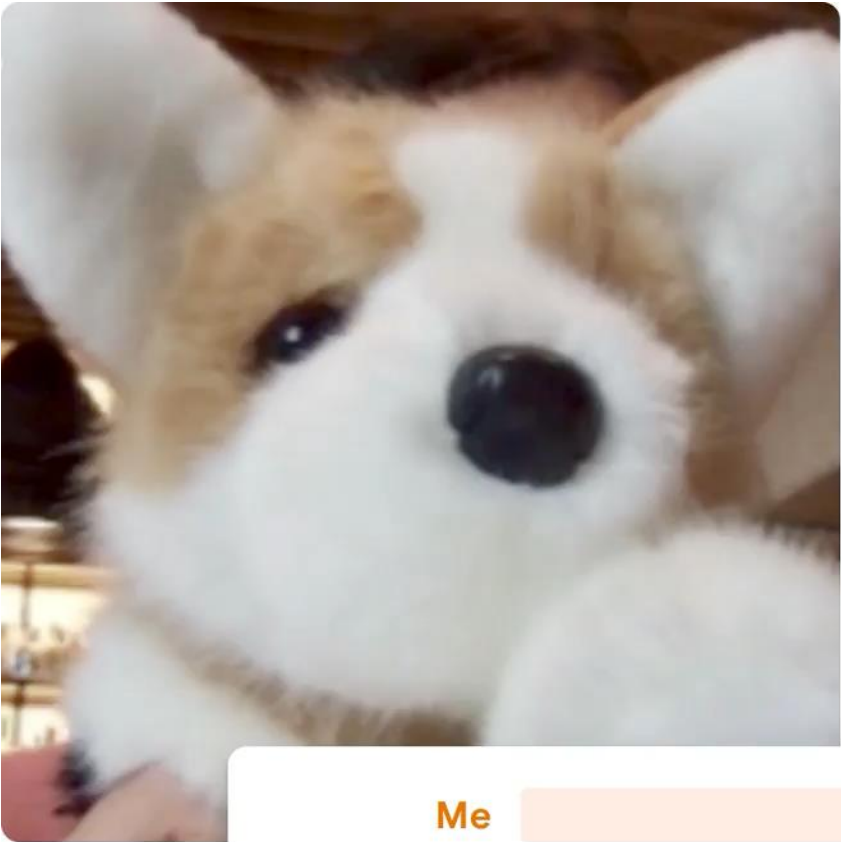
Audio Project

Teach based on one-second-long sounds, from files or your microphone.



Pose Project

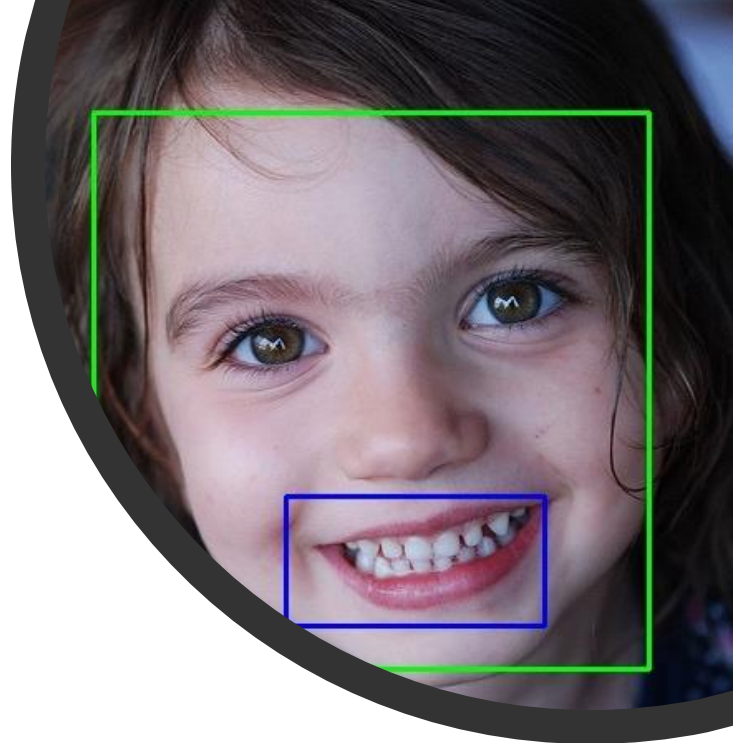
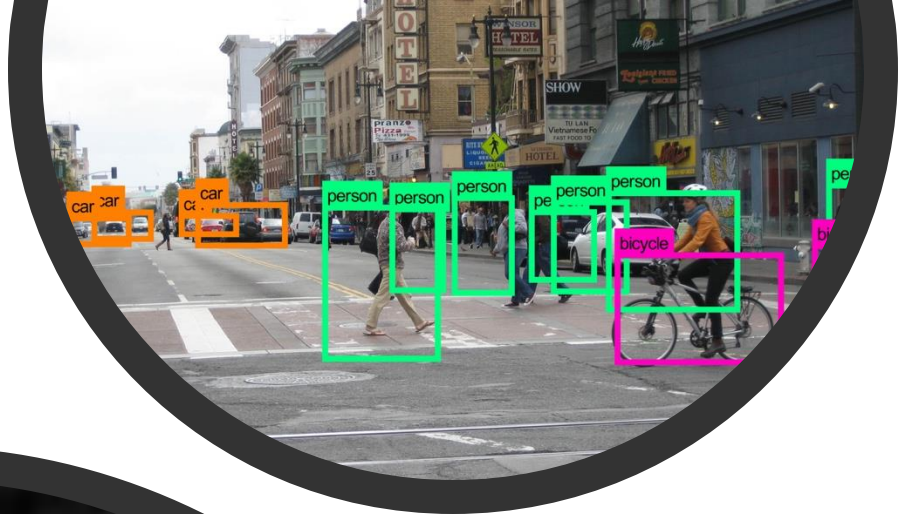
Teach based on images, from files or your webcam.



Me

Me + Dog <3

20%



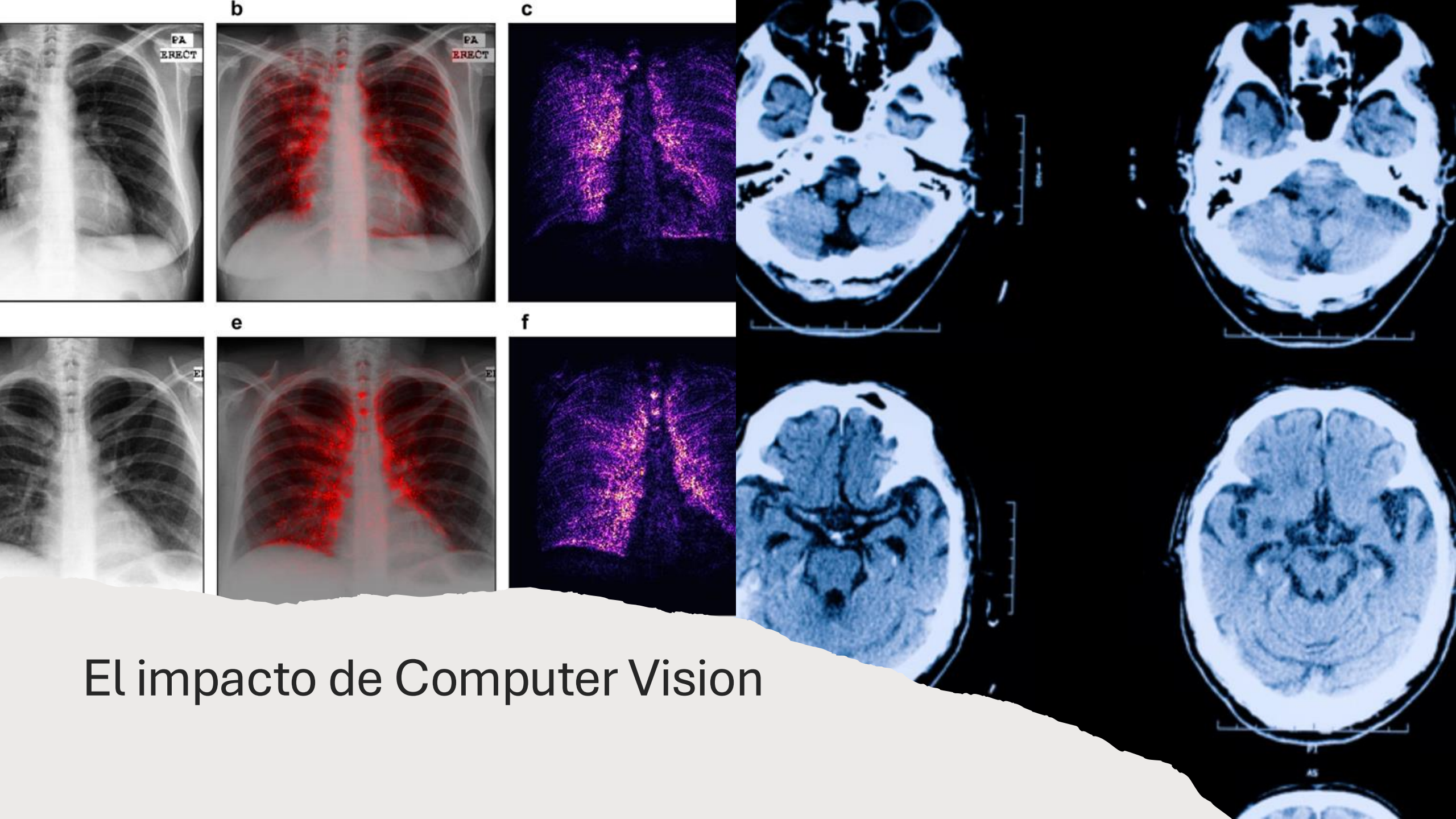
El impacto de Computer Vision





OpenAI Microscope

Open AI Microscope
<https://microscope.openai.com>



El impacto de Computer Vision

RESOURCE | VOLUME 172, ISSUE 5, P1122-1131.E9, FEBRUARY 22, 2018



PDF [4 MB]



Figures



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Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning

[Daniel S. Kermany](#) ¹⁴ • [Michael Goldbaum](#) ¹⁴ • [Wenjia Cai](#) ¹⁴ • ... [M. Anthony Lewis](#) • [Huimin Xia](#) •

[Kang Zhang](#) ¹⁵ • [Show all authors](#) • [Show footnotes](#)

[Open Archive](#) • DOI: <https://doi.org/10.1016/j.cell.2018.02.010> •



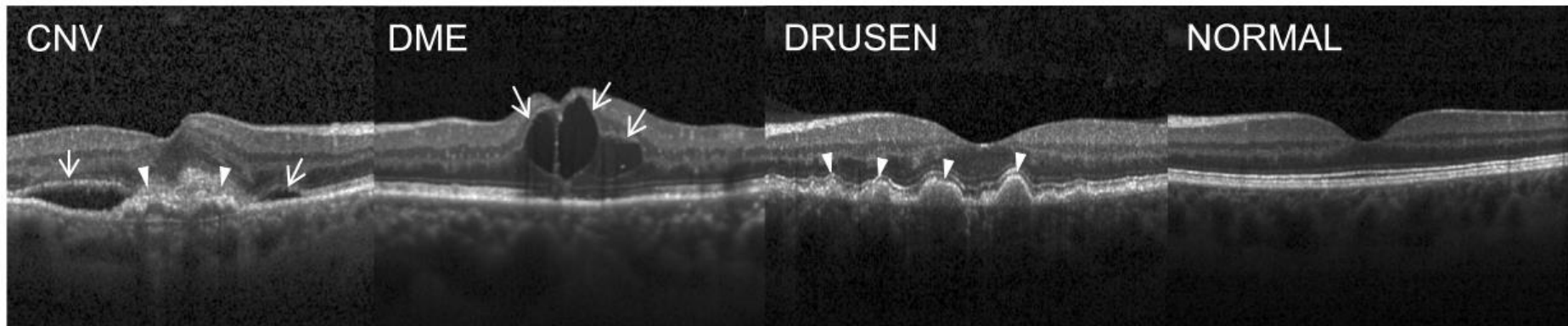
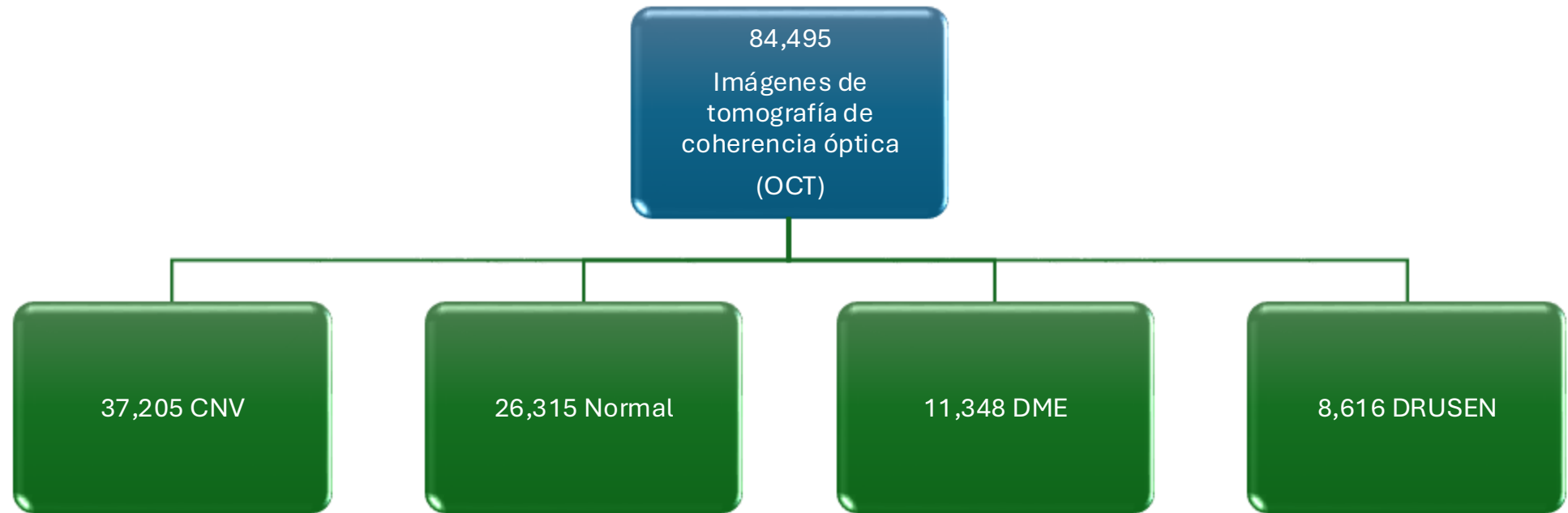
Check for updates



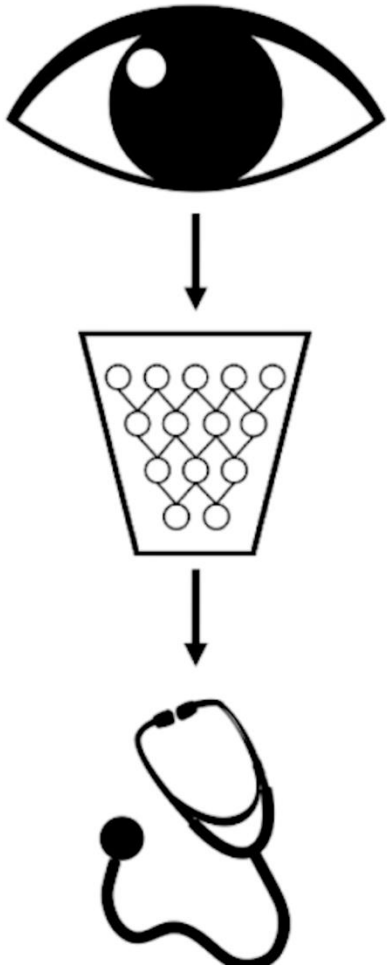
Ejemplo

Ejemplo:

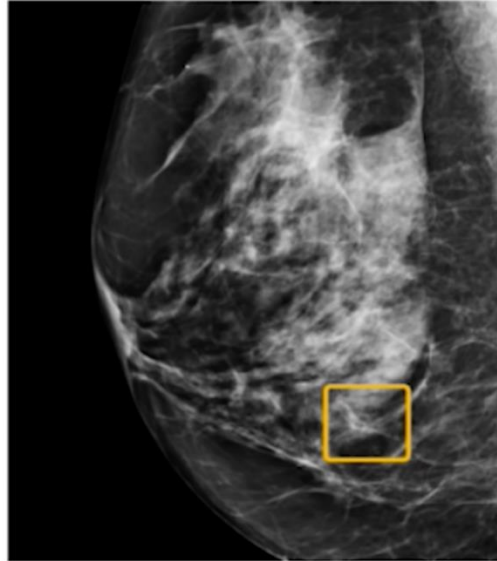
Datos para el estudio (Entrenamiento)



Aplicaciones: Salud



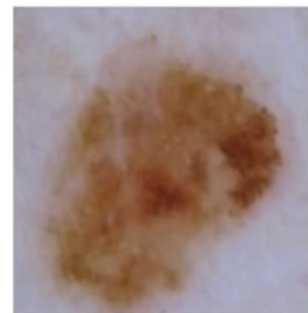
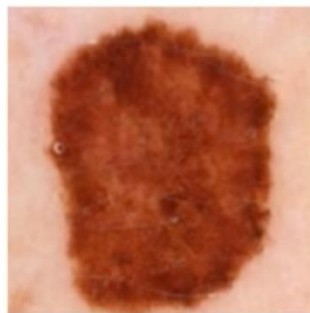
Breast cancer



COVID-19



Skin cancer





Lo que puede ver una computadora



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

What the computer sees

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
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199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
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190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Las imágenes son números



Input Image



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
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190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Pixel Representation



classification

Lincoln

Washington

Jefferson

Obama

$$\begin{bmatrix} 0.8 \\ 0.1 \\ 0.05 \\ 0.05 \end{bmatrix}$$

Tareas en visión computacional

Fuente: Introduction to Deep Learning MIT OpenCourseWare, 2021

Operación de convolución



Extracción de características empleando convoluciones

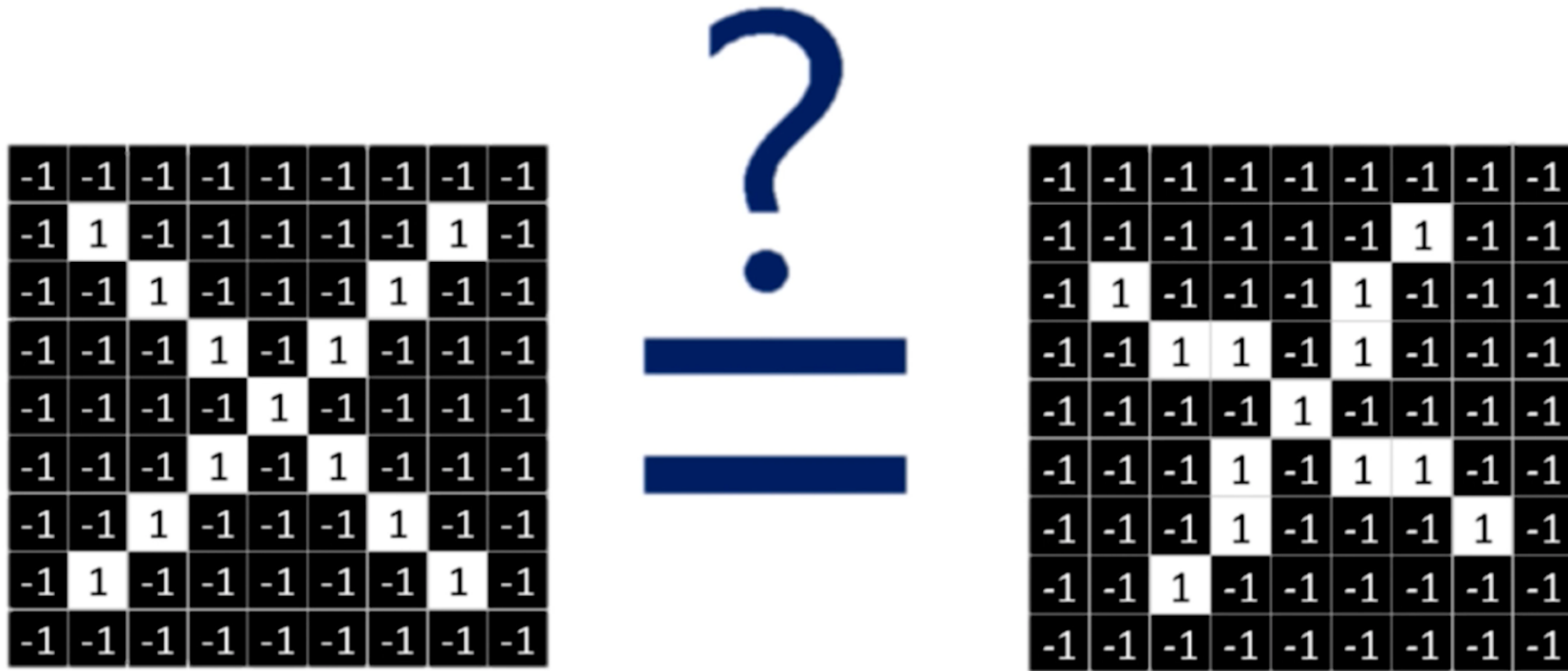
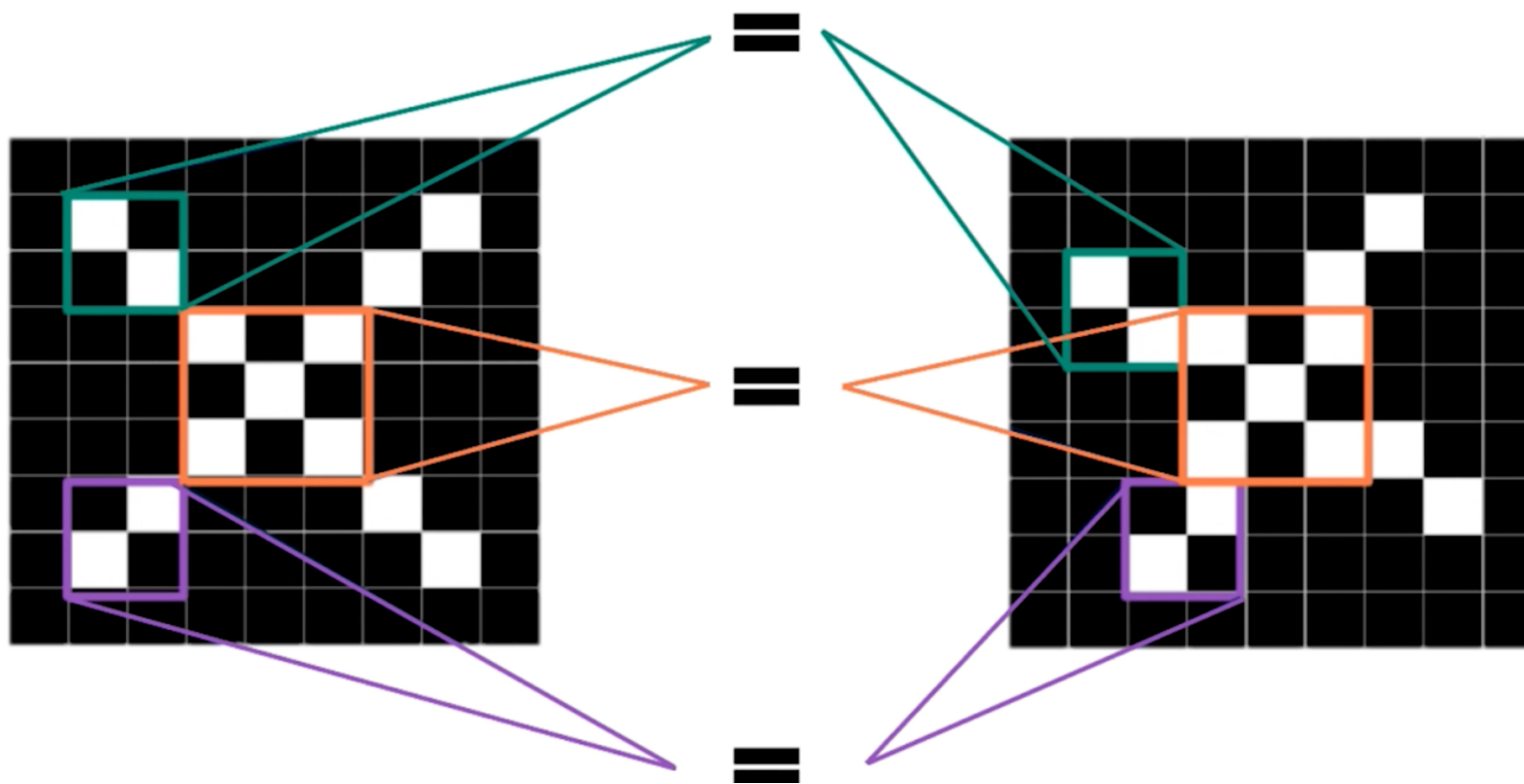


Image is represented as matrix of pixel values... and computers are literal!
We want to be able to classify an X as an X even if it's shifted, shrunk, rotated, deformed.

Características de X



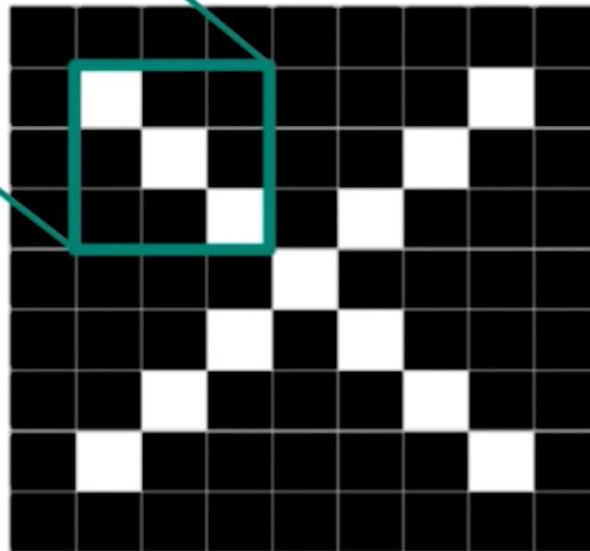
Filtros para detectar las características de X

filters

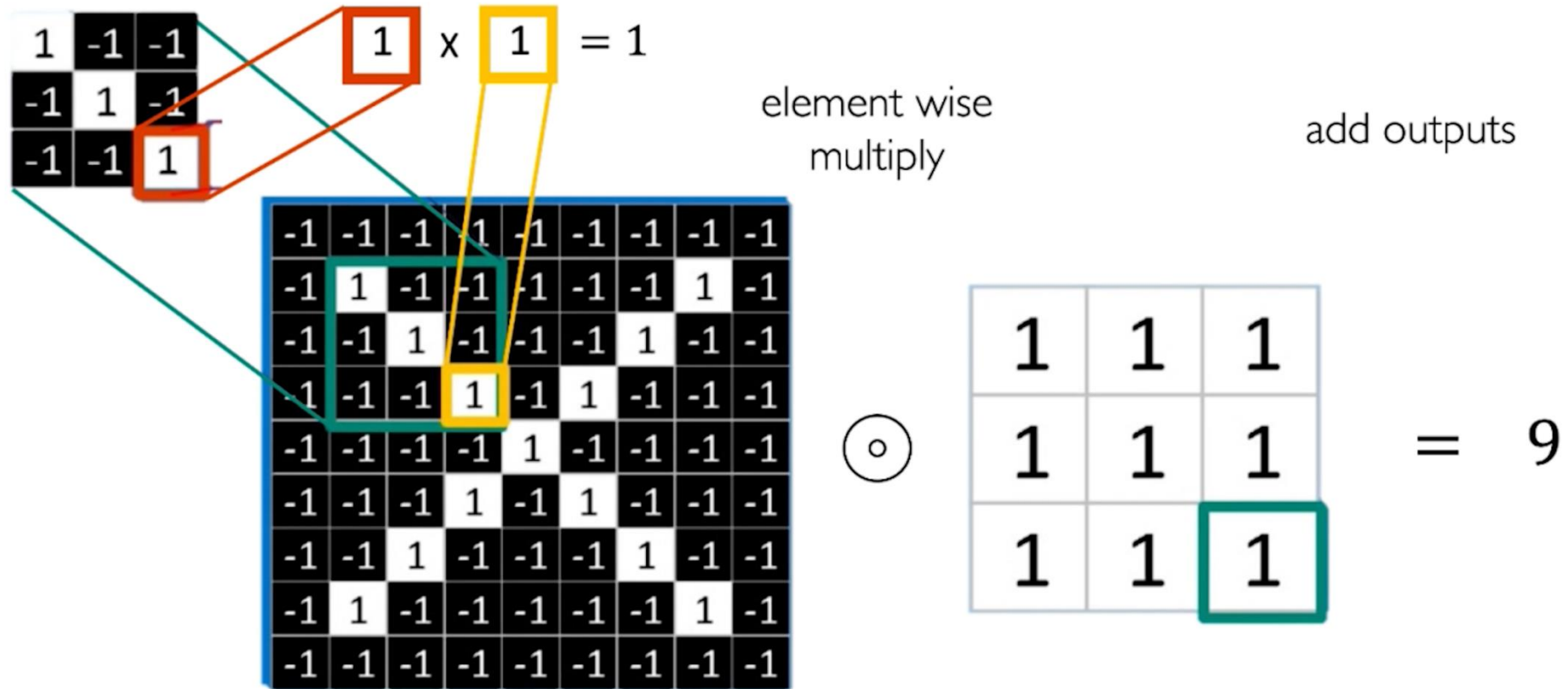
1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1

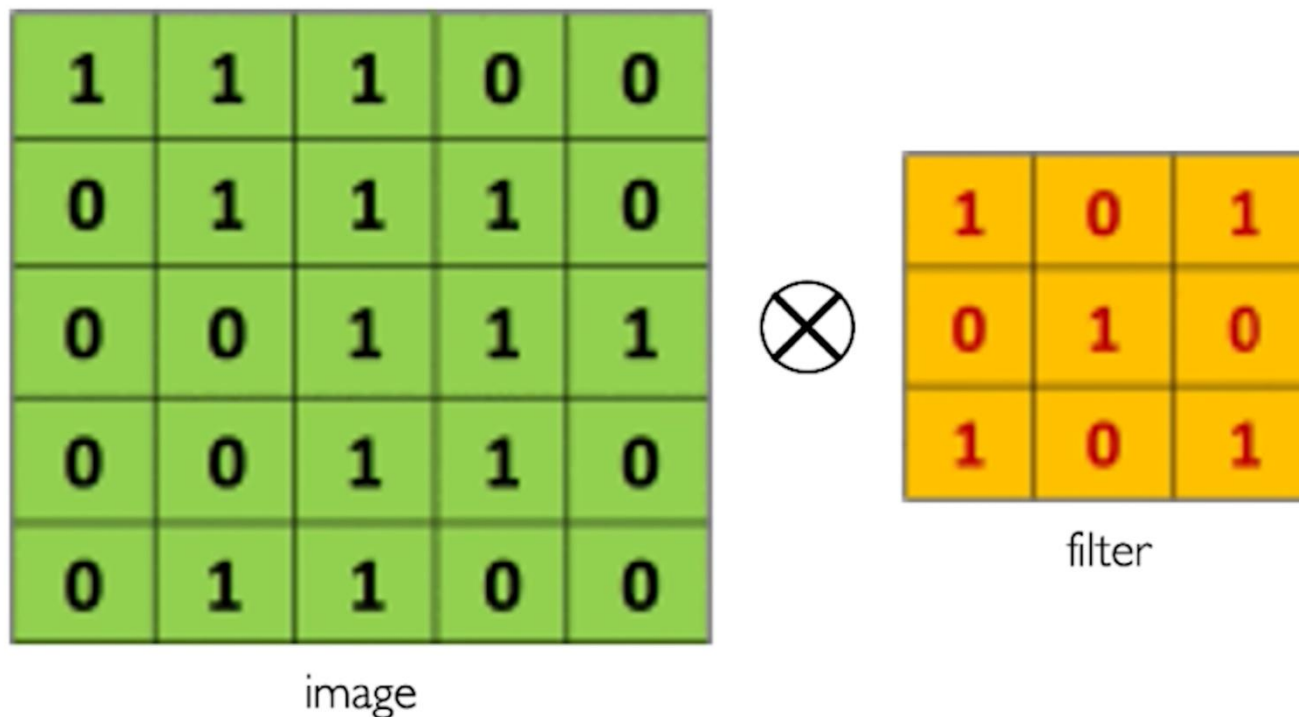


Operación de convolución



Operación de convolución

Suppose we want to compute the convolution of a 5x5 image and a 3x3 filter:



We slide the 3x3 filter over the input image, element-wise multiply, and add the outputs...

Operación de convolución

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0



1	0	1
0	1	0
1	0	1

filter

=

4		

feature map

Operación de convolución

1	1 _{x1}	1 _{x0}	0 _{x1}	0
0	1 _{x0}	1 _{x1}	1 _{x0}	0
0	0 _{x1}	1 _{x0}	1 _{x1}	1
0	0	1	1	0
0	1	1	0	0



1	0	1
0	1	0
1	0	1

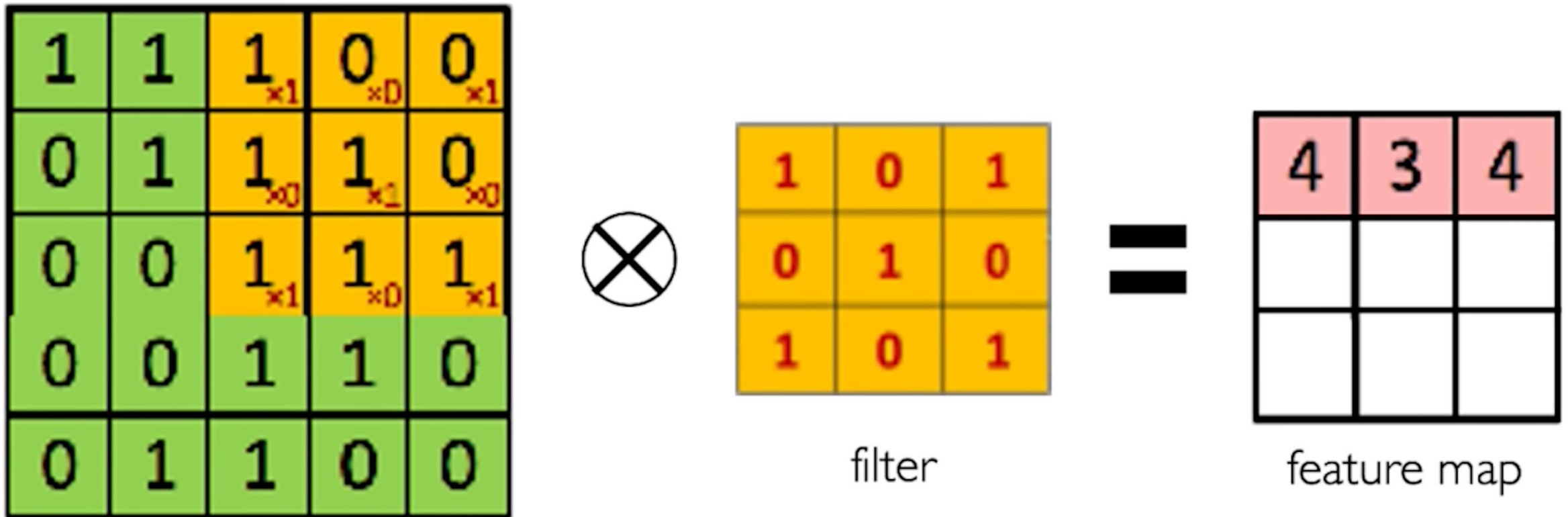
filter



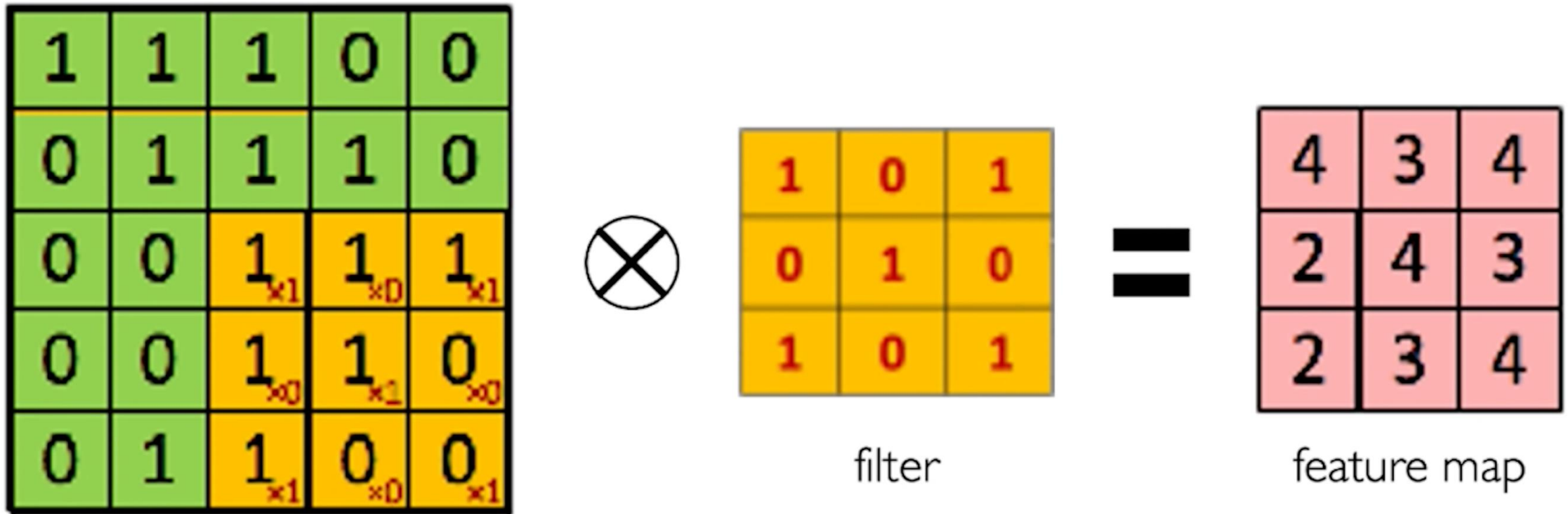
4	3	

feature map

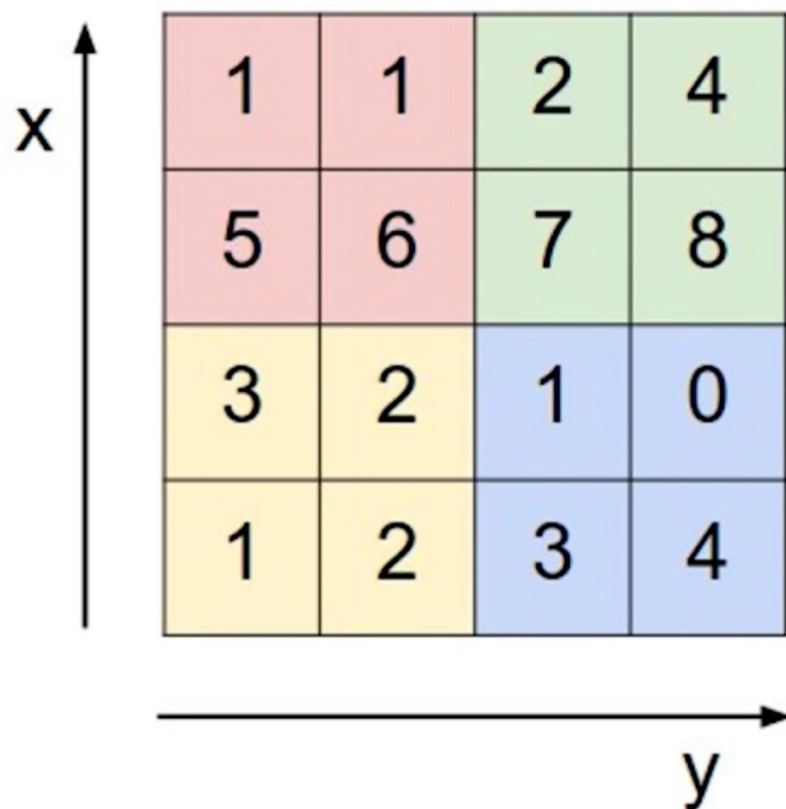
Operación de convolución



Operación de convolución



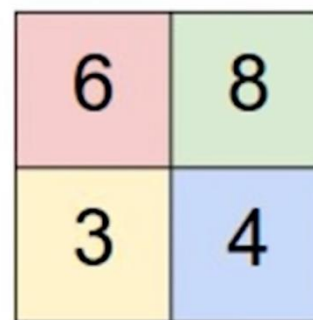
Pooling (agrupación)



1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

max pool with 2x2 filters
and stride 2

```
tf.keras.layers.MaxPool2D(  
    pool_size=(2,2),  
    strides=2  
)
```



6	8
3	4

- 1) Reduced dimensionality
- 2) Spatial invariance

How else can we downsample and preserve spatial invariance?

Convoluciones - Actividad en equipo

- Actividad:
 - Experimentar aplicando Kernels de convolución a imágenes.
 - <https://setosa.io/ev/image-kernels/>

Image Kernels

Explained Visually

Post Me gusta 440 Compara

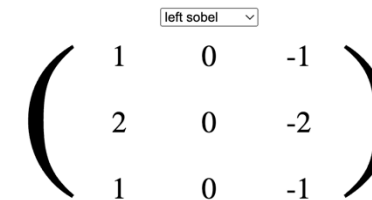
By Victor Powell

An image kernel is a small matrix used to apply effects like the ones you might find in Photoshop or Gimp, such as blurring, sharpening, outlining or embossing. They're also used in machine learning for 'feature extraction', a technique for determining the most important portions of an image. In this context the process is referred to more generally as "convolution" (see: [convolutional neural networks](#).)

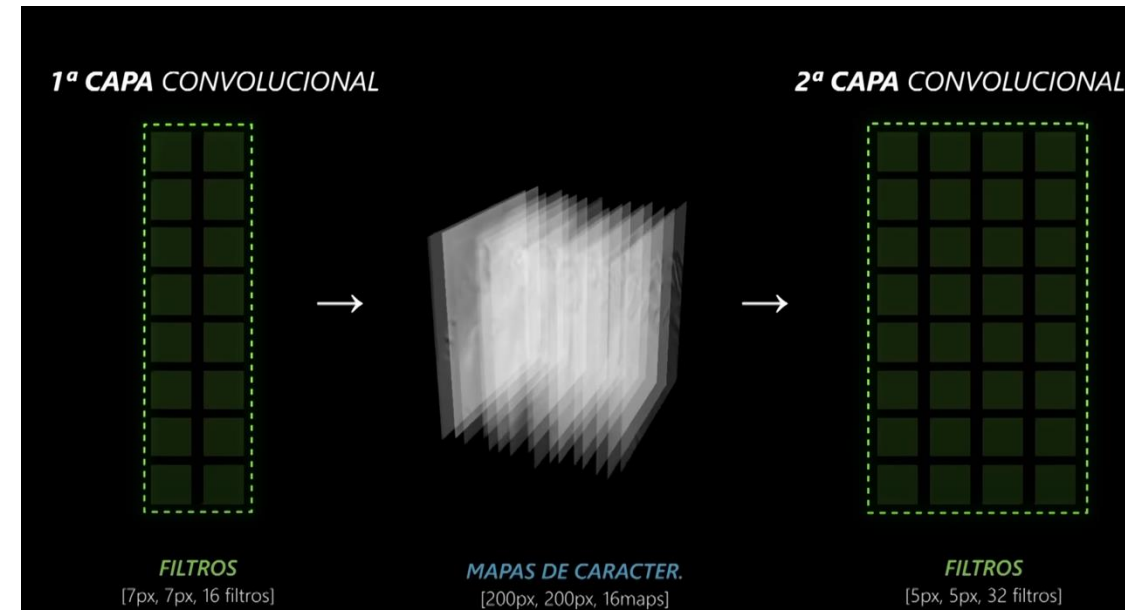
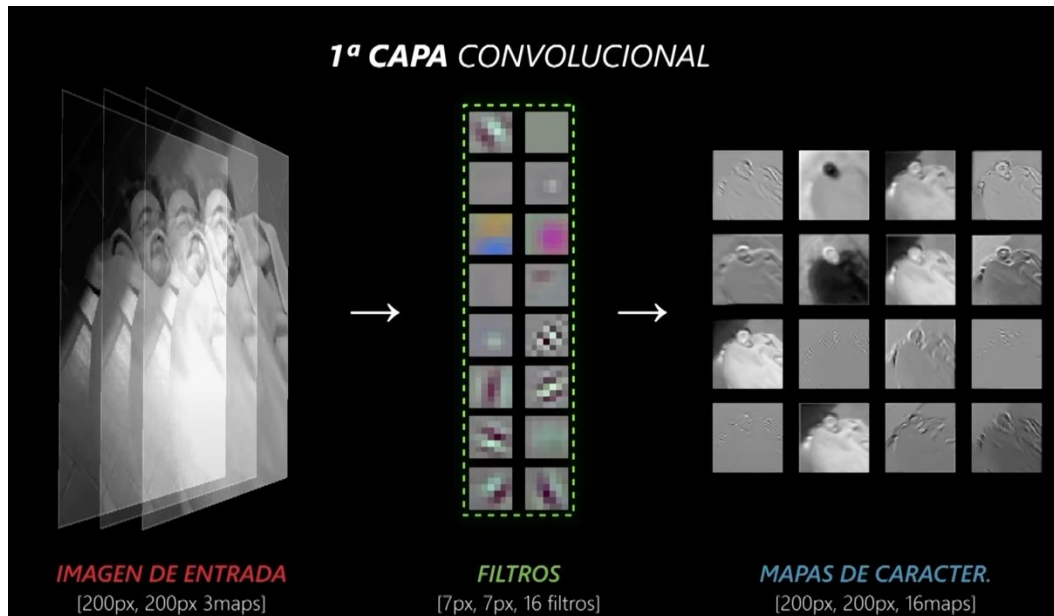
To see how they work, let's start by inspecting a black and white image. The matrix on the left contains numbers, between 0 and 255, which each correspond to the brightness of one pixel in a picture of a face. The large, granulated picture has been blown up to make it easier to see; the last image is the "real" size.



Let's walk through applying the following 3x3 **left sobel** kernel to the image of a face from above.



Red Neuronal Convolucional

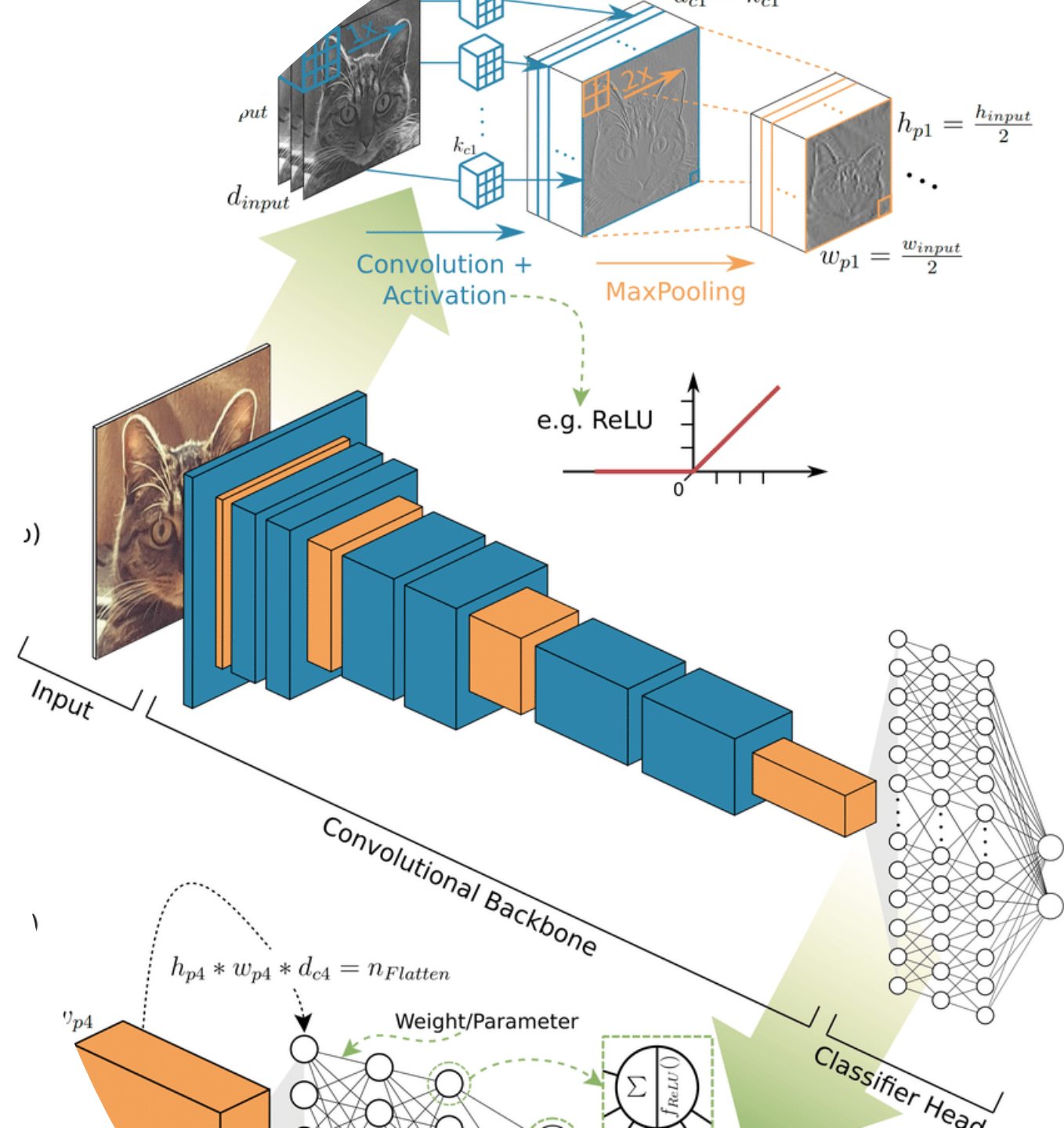


Convoluciones y Deep Learning



Redes Neuronales Convolucionales (CNNs)

Deep Learning



CNNs para clasificación

1. Convolución:

Aplicar filtros para generar mapas de características

2. Introducir no linealidad (función de activación):

Usualmente se utiliza la función RELU

3. Agrupación:

Agrupar los datos de cada mapa de características.



```
tf.keras.layers.Conv2D
```

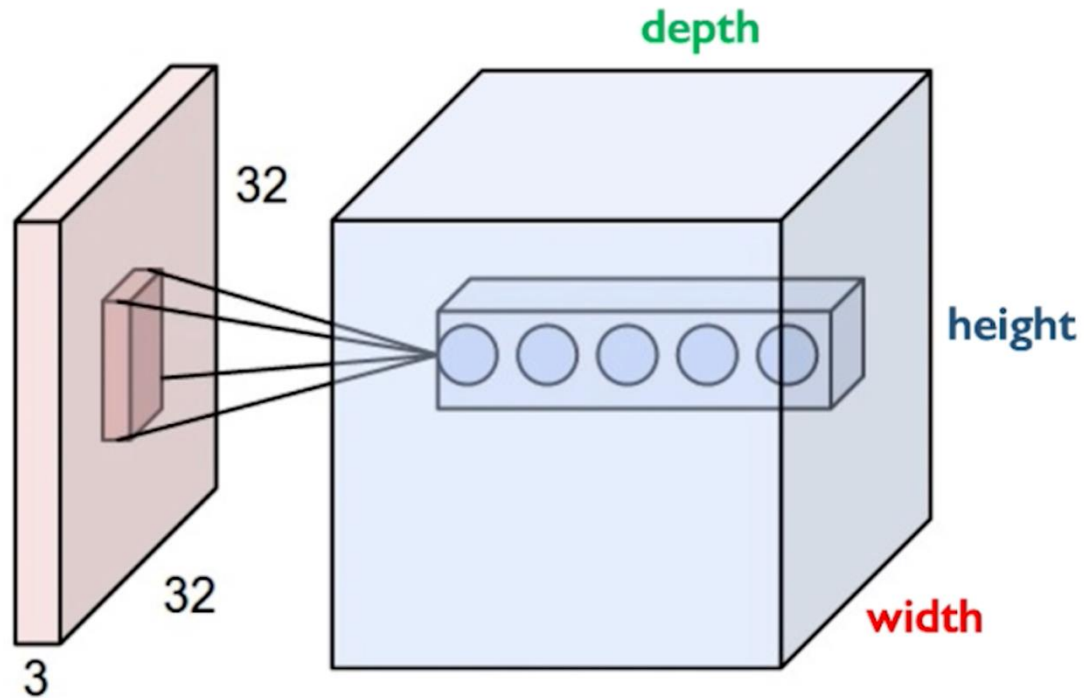


```
tf.keras.activations.*
```



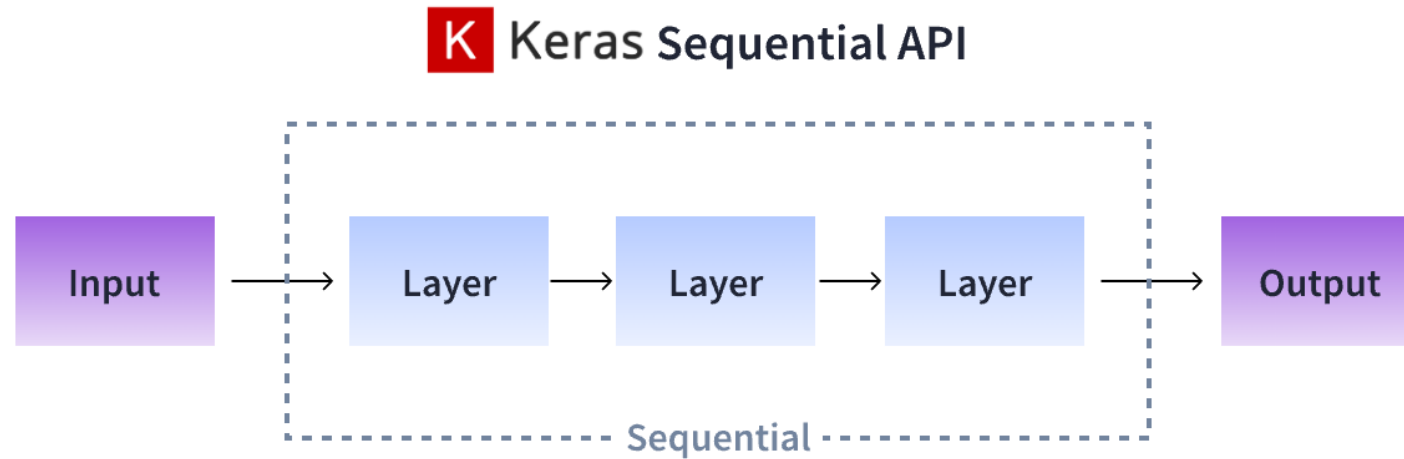
```
tf.keras.layers.MaxPool2D
```

CNNs: Arreglo espacial salidas



```
tf.keras.layers.Conv2D( filters=d, kernel_size=(h,w), strides=s )
```

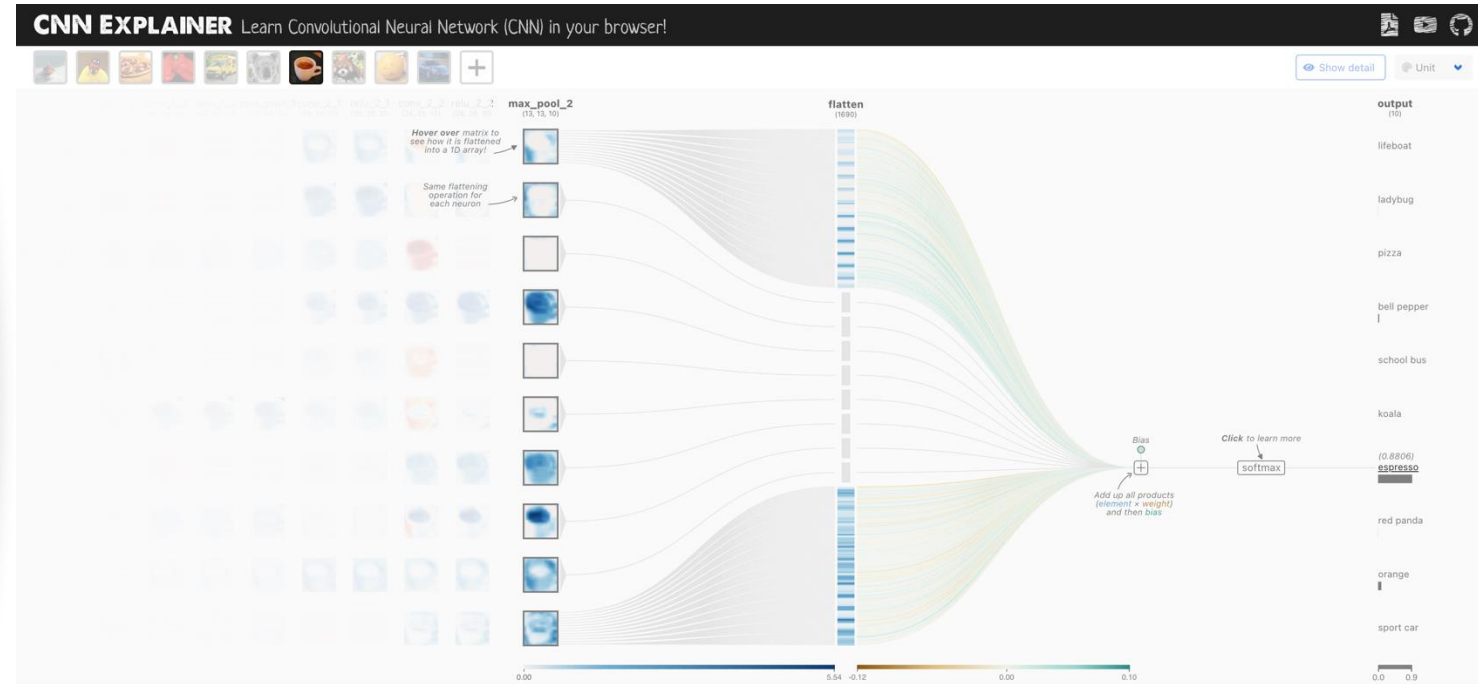
Apilando capas Convolucionales



```
1 model = models.Sequential()
2 model.add(layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)))
3 model.add(layers.MaxPooling2D((2, 2)))
4 model.add(layers.Conv2D(64, (3, 3), activation='relu'))
5 model.add(layers.MaxPooling2D((2, 2)))
6 model.add(layers.Conv2D(64, (3, 3), activation='relu'))
7
8 # Añadiendo capas densas en el final
9 model.add(layers.Flatten())
10 model.add(layers.Dense(64, activation='relu'))
11 model.add(layers.Dense(10))
```

CNN Explainer

<https://poloclub.github.io/cnn-explainer/>



What is a Convolutional Neural Network?

CNNs para clasificación



```
tf.keras.layers.Conv2D
```



```
tf.keras.activations.*
```



```
tf.keras.layers.MaxPool2D
```

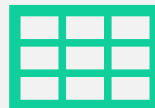


1. Convolución: Aplicar filtros para generar mapas de características



2. Introducir no linealidad (función de activación):

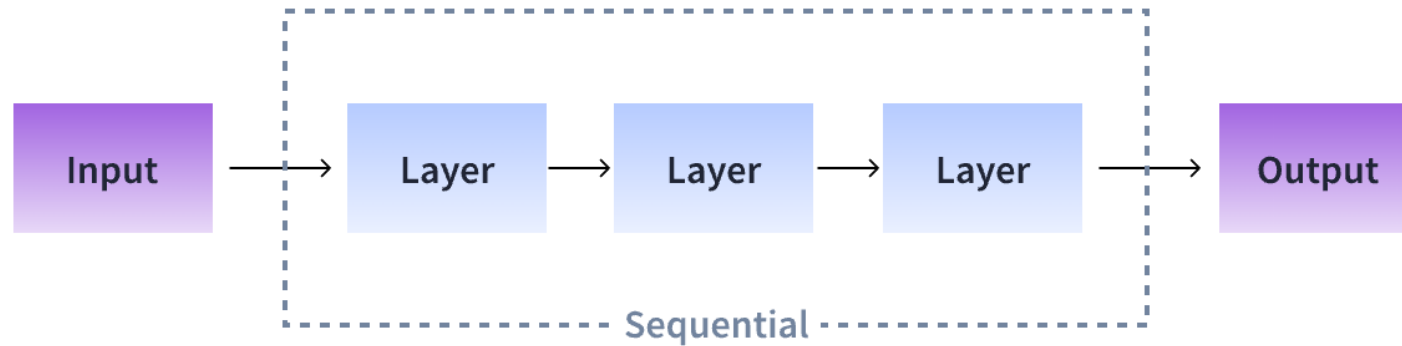
Usualmente se utiliza la función RELU



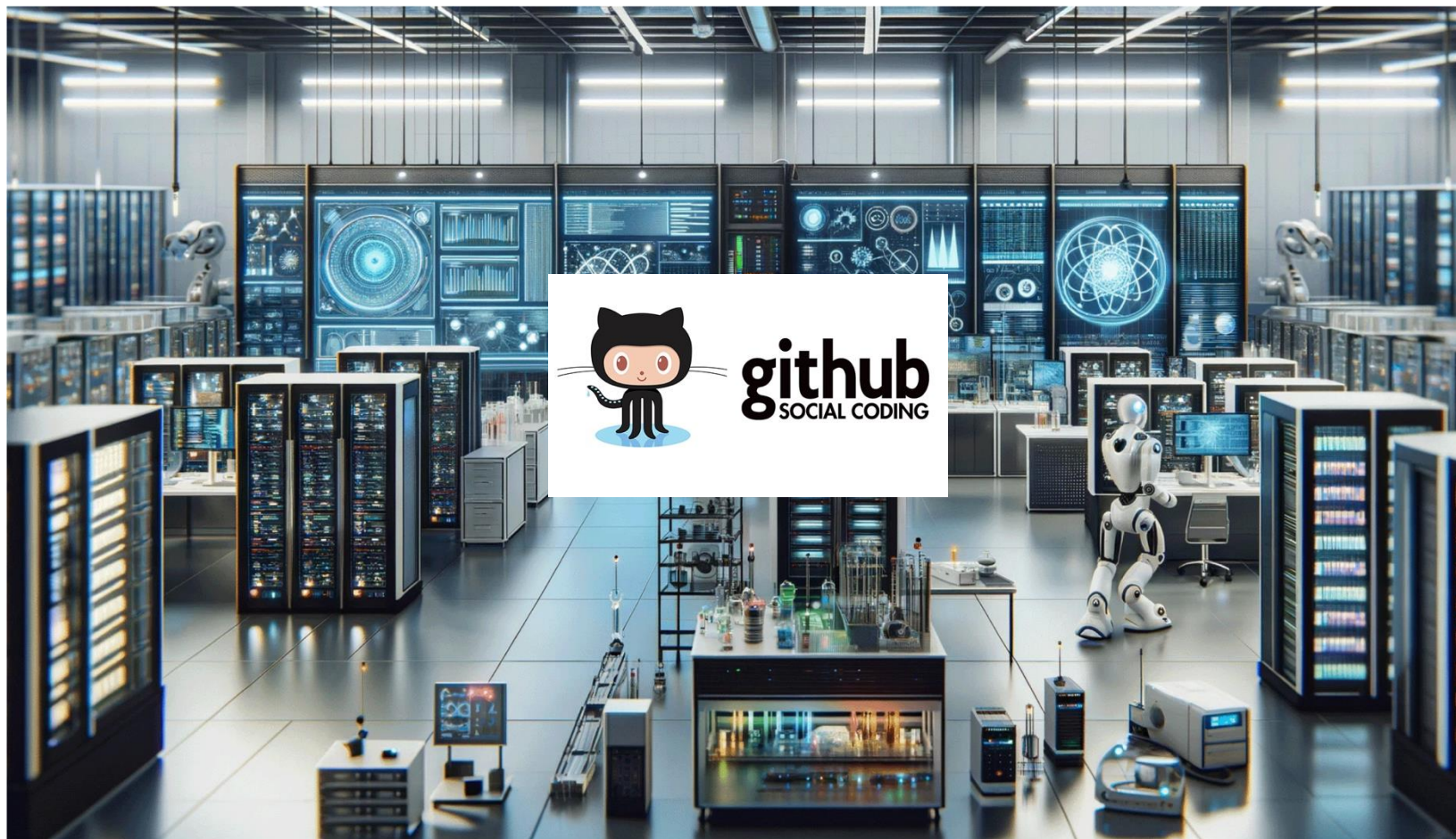
3. Agrupación (Pooling): Agrupar los datos de cada mapa de características.

Apilando capas Convolucionales

 Keras Sequential API



```
1 model = models.Sequential()
2 model.add(layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)))
3 model.add(layers.MaxPooling2D((2, 2)))
4 model.add(layers.Conv2D(64, (3, 3), activation='relu'))
5 model.add(layers.MaxPooling2D((2, 2)))
6 model.add(layers.Conv2D(64, (3, 3), activation='relu'))
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8 # Añadiendo capas densas en el final
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10 model.add(layers.Dense(64, activation='relu'))
11 model.add(layers.Dense(10))
```





TensorFlow

Demo Base: CNN

y CIFAR 10

CIFAR - 10

airplane

automobile

bird

cat

deer

dog

frog

horse

ship

truck

